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Demand Line

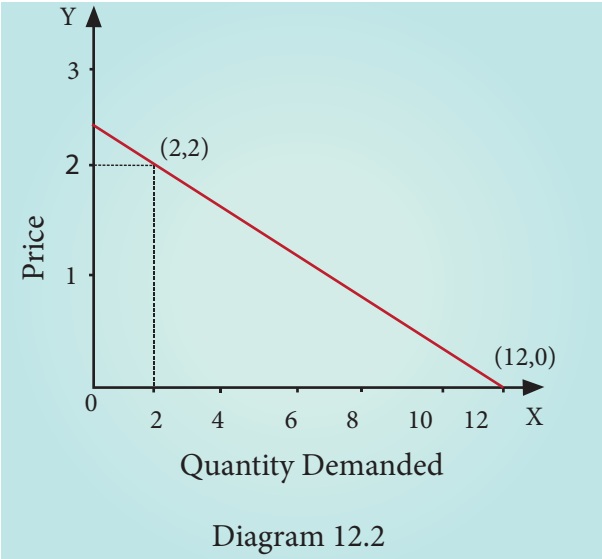


Diagram 12.2

Supply Line

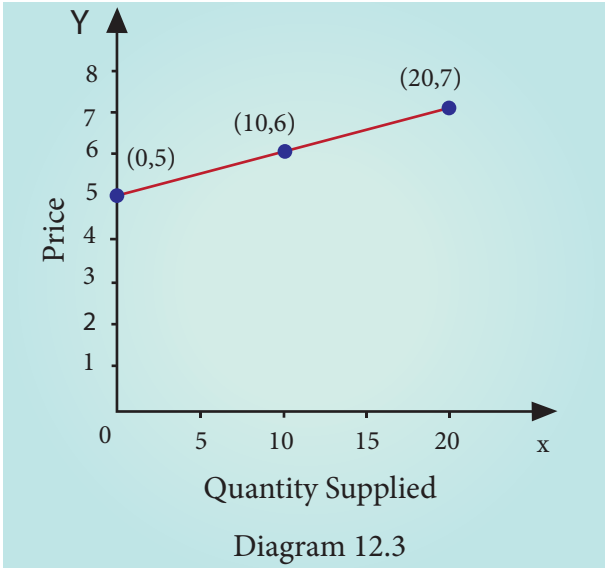


Diagram 12.3

Price-quantity relationship is negative in demand function. $Q_d = 12 - 5X$ or $Q_d = 12 - 5P$. If $P = 2$, $Q_d = 2$.

When P assumes 0, only 12 alone remains in the equation. This is called Intercept or Constant, if $P = 0$ and $Q_d = 12$.

In Marshallian analysis, money terms measured in Y-axis and physical units are measured in X-axis. Accordingly, price is measured in Y-axis and quantity demanded is measured in X-axis

Example: 12.2

Find the supply function of a commodity such that the quantity supplied is zero, when the price is ₹5 or below and the supply (quantity) increases continuously at the constant rate of 10 units for each one rupee rise when the price is above ₹5.

Solution:

To construct the linear supply function atleast two points are needed. First data point of supply function is obtained from the statement that the quantity supplied is zero, when the price is ₹5, that is (0, 5). The second and third data points of supply function can be obtained from the statement that supply increases 10 units for each one rupee rise in price, that is (10, 6) & (20, 7).

When $p = 5$, supply is zero. When $p = 6$, supply is 10 and so on. When p is less than 5, say 4, supply is -10, which is possible in mathematics. But it is meaningless in Economics. Normally supply curve originates from zero, noting that when price is zero, supply is also zero.

The equation of the straight line joining two data points (10, 6) and (20, 7) is given as

The equation of the straight line is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

substituting the values of (x_1, y_1) (x_2, y_2) by (10, 6) , (20, 7) respectively,

$$\frac{y - 6}{7 - 6} = \frac{x - 10}{20 - 10}$$

$$\frac{y - 6}{1} = \frac{x - 10}{10}$$